Soham Sadavarte

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Education

International Institute of Information Technology, Hyderabad *Aug 2023 – Present*
B.Tech in Computer Science and Engineering (CGPA: 8.60/10.0)

RMG Maheswari School, Surat *Apr 2021 – Mar 2023*
12th Board: (CBSE) Grade: 93.4%

Delhi Public School Tapi, Surat *Apr 2019 – Mar 2021*
10th Board: (CBSE) Grade: 95.8%

Coursework

Data Structures & Algorithms | **Operating Systems & Networks** | **Algorithm Analysis & Design**
Design & Analysis of Software Systems | **Machine Data & Learning** | **Probability & Statistics**
Introduction to Information & Security | **Data & Applications** | **Linear Algebra**

Achievements

JEE Mains 2023: AIR 889 (Top 0.07%)
JEE Advanced 2023: AIR 3189 (Top 1.6%)
KVPY 2021: AIR 2000 (Merit List)
Codeforces Rating: 1433 (sohamsada123)

Experience

Team Project Engineer, Virtual Labs, IIIT Hyderabad *Jan 2025 – Apr 2025*

- Collaborated as a team member to build an **agentic pipeline** using agentic frameworks to streamline code generation.
- Enabled **automated experiment creation using 6 coordinated agents**, improving reproducibility .

Projects

NFS—Network File System — *C, TCP sockets, multi-threading* GitHub

- Created a distributed file system that enables **efficient client read/write** via a centralized naming server. Also has **fault tolerant** file replication system with **replication factor of 3**.
- Directed **TCP communication with multithreaded** client handling for **8+ concurrent clients** and **LRU caching** for optimal file access.
- Integrated automated server registration (up to 4–5 machines), directory listing, metadata retrieval, and **inter-server file transfers** up to 1 GB.

AKS PHOTOSHOP—Photo Slideshow Web App — *Python, Flask, OpenCV, SQL, CockroachDB* GitHub

- Composed a full-stack application that converts image sequences into music-synced videos, processing up to **25+ images** per slideshow with **customizable image timing**.
- Enabled image upload, smooth transitions, audio mixing with **5 tracks**, and a **real-time preview**.

Hospital Management System (Relational DB Project) — *Python, ER Modeling, SQL* GitHub

- Conceived and authored a **hospital database mini-world** using ER modeling and SQL, featuring **10+ entities** with comprehensive schema design.
- Optimized data by **normalizing to 2NF**, maintaining integrity across 500+ records to enable efficient querying and secure control access.

Buy-Sell @ IIITH (MERN Stack Web App) — *MERN Stack, JWT, Bcrypt* GitHub

- Built a **MERN stack e-commerce portal** with JWT authentication, hashed passwords, and secured routes including 5+ protected user routes.
- Added features like category and keyword search, adding products and **handling 100+ products**, filtering, shopping cart, OTP-based ordering, and role-based views.

Linux Shell — *C, POSIX signals* GitHub

- Produced a custom bash-like shell in C with **15+ features** like semicolon-separated commands, I/O redirection, piping, background and foreground process management, signal handling and command history.
- Implemented **recursive file search** with **3 filtering flags (-d,-e,-f)** and a **TCP socket-based** manual page fetcher that parses and **removes HTML tags** for **clean CLI output** and seamless user experience.

Technical Skills

Programming Languages: C++, C, Python, JavaScript, HTML, CSS, Assembly.
Technologies: MySQL, Flask, MERN Stack, MongoDB Atlas, Git